

Umut Ekin Demirkol

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EDUCATION

Yıldız Technical University

B.Sc. in Control and Automation Engineering (100% English)

Sept. 2022 – Present

Istanbul, Türkiye

EXPERIENCE

GDS Engineering R&D

R&D and Software Engineer

Jul. 2026 – Present

Istanbul, Türkiye

- Lead developer of the TÜBİTAK maritime energy efficiency project; built the physics engine of a real-time ship digital twin in Python against measured marine engine test data, streamed to a LabVIEW operator interface over UDP, and the voyage optimization layer scoring candidate routes on speed, fuel consumption and emissions
- Sole developer of a real-time engine sound module in Python that generates the audio signal directly from live simulation state

Özdaş Mühendislik

Automation Engineer Intern

Feb. 2026

Istanbul, Türkiye

- Built and wired industrial control panels, bridging theoretical control logic and physical application
- Applied CiA 402 homing methods 23, 27 and 34 for mechanical referencing and optimized dynamic performance via servo drive and motor tuning using CODESYS over EtherCAT

Beijer Electronics

Automation Engineer Intern

Jun. 2025 – Sept. 2025

Istanbul, Türkiye

- Programmed in CODESYS (IEC 61131-3) and designed HMI screens for the X2 Base series in iX Developer
- Integrated industrial architectures via OPC UA, Modbus TCP/IP and BACnet, validating behavior in system simulation
- Documented recurring HMI faults and their fixes into a troubleshooting reference published on Beijer's own forum, reaching 3,000+ users and shared with the sales team

PROJECTS

Vision-Based Detection & Control System

Current

- Trained a YOLOv8-Nano model (~6 MB) on a restructured two-class dataset for real-time inference; built a Python/OpenCV pipeline pulling the live camera feed over RTSP and running detection frame by frame on the host machine
- Modeling quadrotor flight dynamics in MATLAB & Simulink, comparing PID, nonlinear PID and LQR for stable, energy-optimal flight

Smart Factory Automation & SCADA System

Current

- Developing CODESYS control logic validated against a Factory I/O virtual production line, with an Ignition SCADA/HMI layer linked over MQTT and Modbus TCP/IP

Smart Home Automation & Energy Management System

Jan. – Jul. 2026

In collaboration with Prof. Bernard Riera | Supported by Beijer Electronics

- Project Leader of an international collaboration: implemented industrial-grade CODESYS logic managing 400+ I/O points, interfaced with HOME I/O simulation for scenario validation
- Designed the interface in iX Developer plus a custom web platform, with real-time exchange over Modbus TCP/IP and MQTT for energy consumption monitoring and optimization

SKILLS & CERTIFICATIONS

Software & Tools: CODESYS (IEC 61131-3), Python, C, MATLAB & Simulink, LabVIEW, iX Developer, Ignition SCADA, Factory I/O & HOME I/O, OpenCV & YOLO

Protocols & Hardware: OPC UA, Modbus TCP/IP, EtherCAT, BACnet, MQTT, UDP, PLC & HMI Hardware, Servo Drives, ESP32

Courses: Universal Robots Academy e-Series Core & Pro Track; LabVIEW Essentials (Udemy); CODESYS Fundamentals & Simulation (Udemy); Smart Relay & Micro-Level PLC Programming, 76h (Elginkan Foundation); McKinsey Forward Program

Languages: Turkish (Native), English (C1 – CEFR)

VOLUNTEERING & EXTRACURRICULAR ACTIVITIES

CIGRE YTU Student Chapter – Board Member & General Director

2023 – 2026

- Oversaw strategic leadership and corporate collaborations for the chapter after progressing from active member
- Lead Organizer of the Yıldız Energy Summit: secured its largest sponsor and managed on-site team logistics

TÜREK Conference – Organization Team Member

2024 – 2025

- Two-year student volunteer providing logistical and technical support for exhibiting companies at conference booths